# OpenWebSearch.eu — A European Open Web Index and its potential Applications

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GI AK-OSS Themenabend · 2026-05-04



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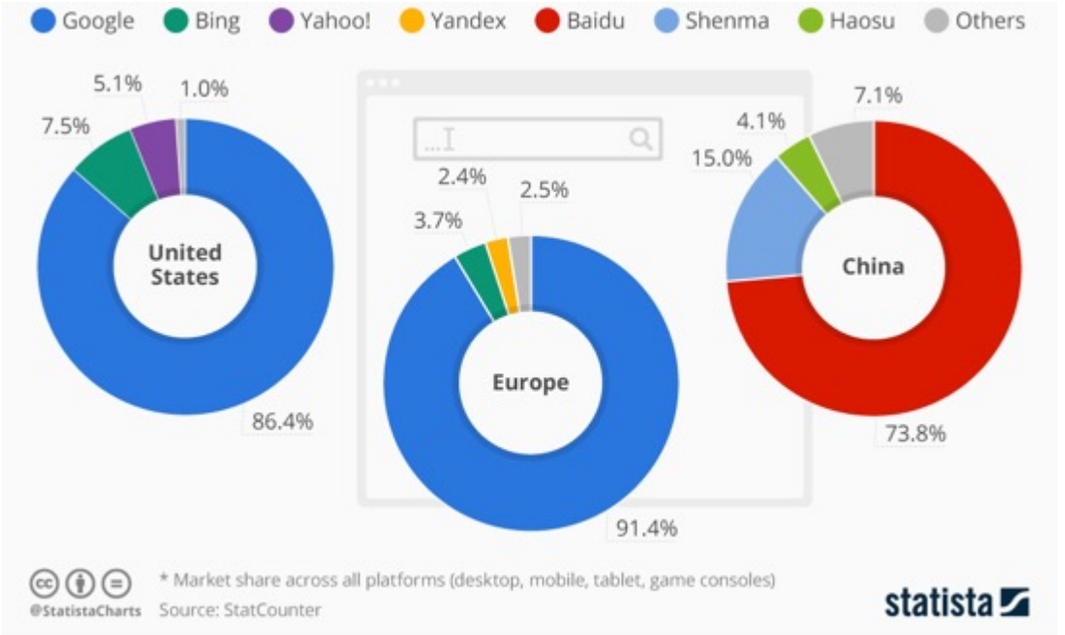
Bayerisches Staatsministerium für Wirtschaft, Landesentwicklung und Energie

# Web Search Today

A critical resource managed as an oligopoly

# Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



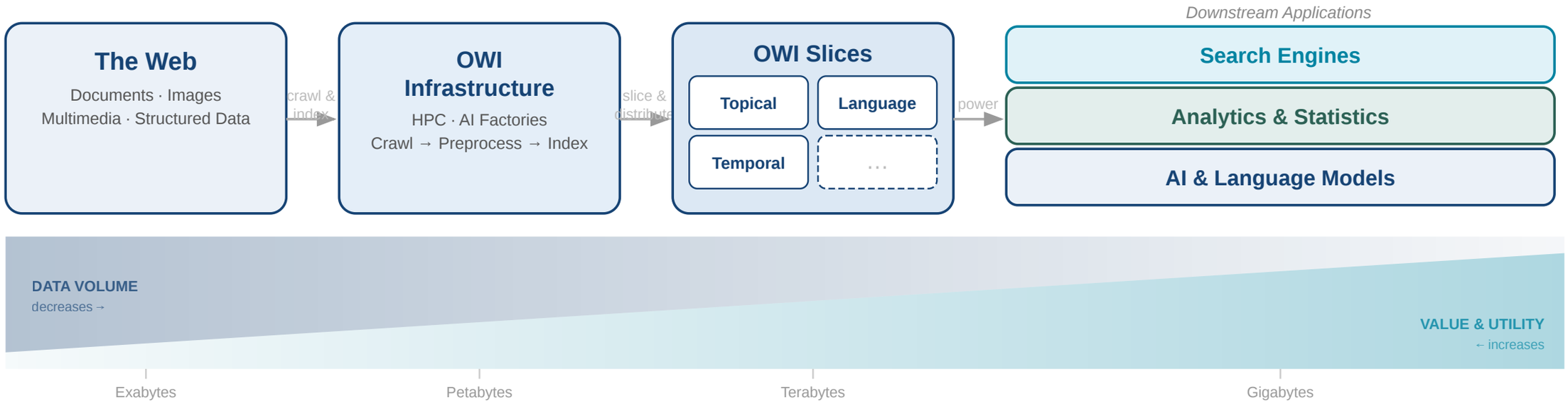

Plus 16 financially supported third-party projects (FSTPss)

Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe




*Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).*

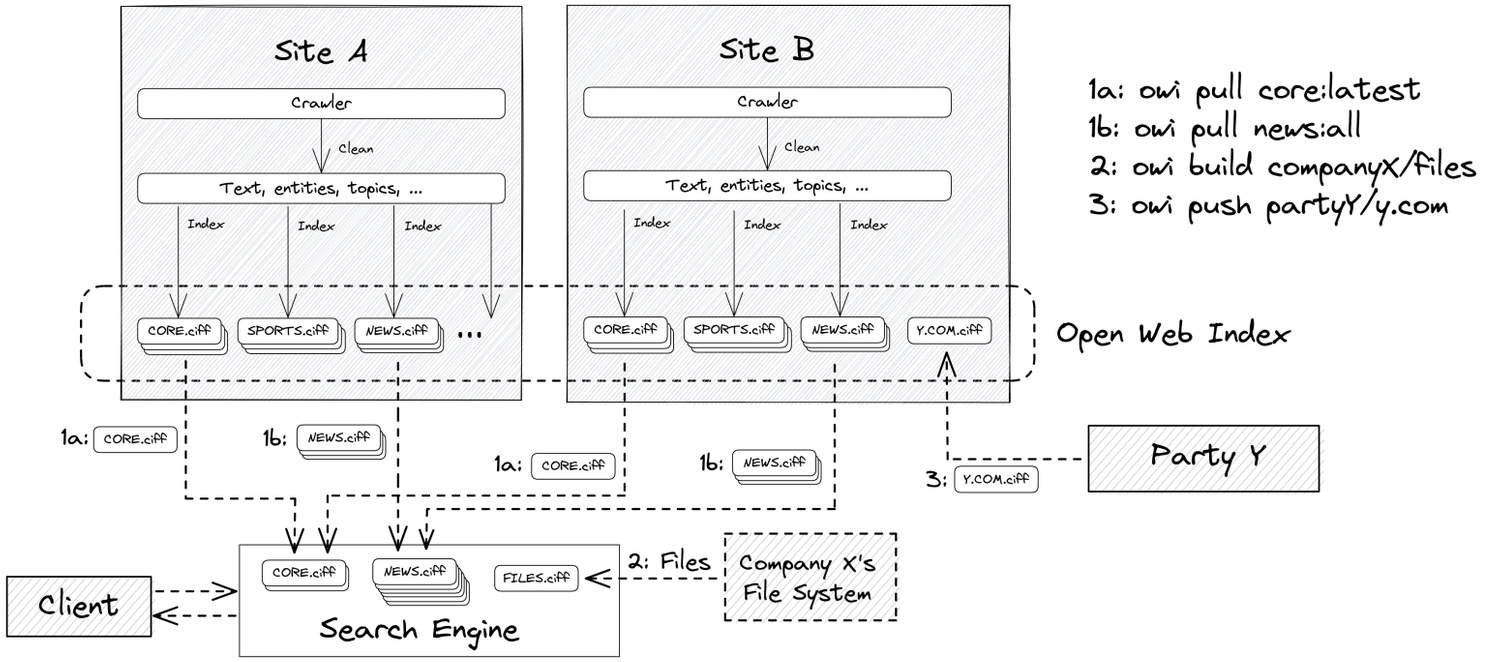
# What is the Open Web Index?

# The Open Web Index

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### Federated production

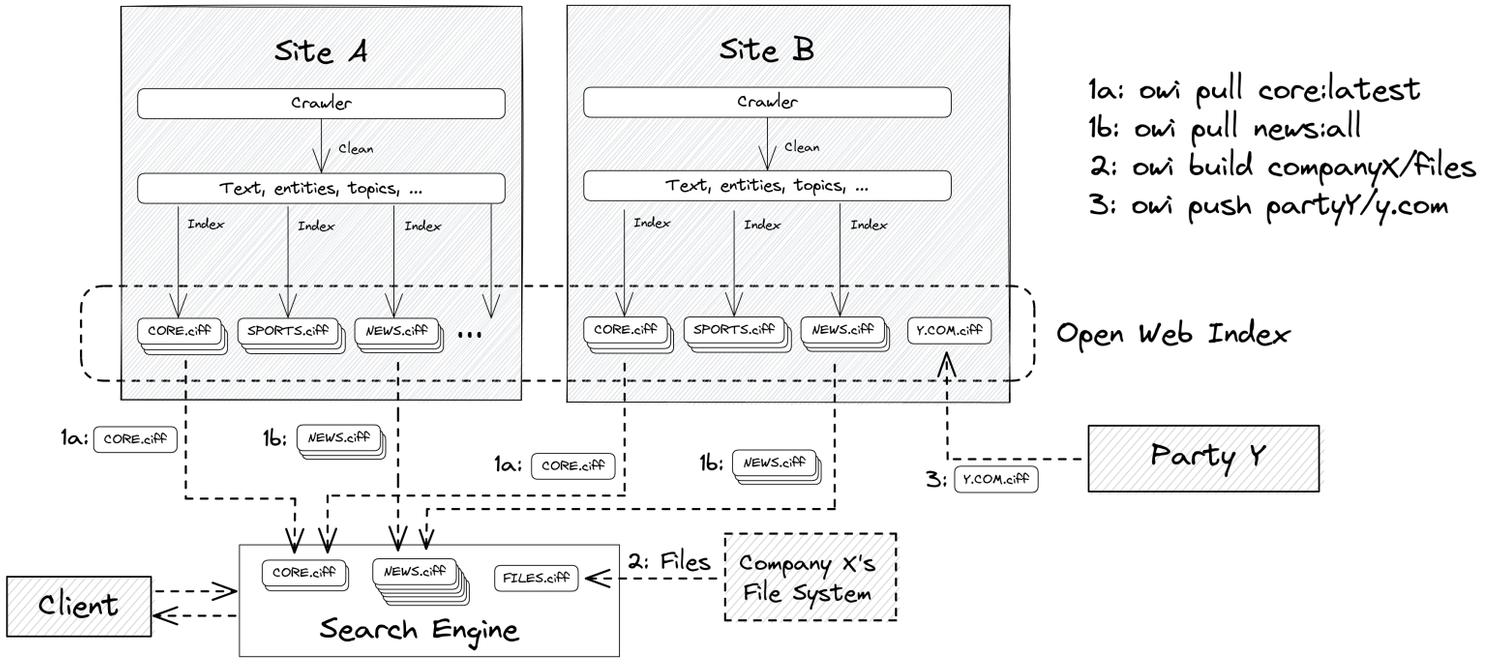
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

### Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

# The Open Web Index

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### Three output types per daily run — the building blocks of a shard:

- `index.ciff.gz` — sparse inverted index (CIFF format; pre-configured tokenizer)
- `metadata_*.parquet` — document metadata & enrichments (topics, microdata, locations, language, etc.; extensible)
- Dense embeddings (GPU-enriched slice, Jina V3) for AI-based search and classification (on selected subsets)

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

| | |
|---|---|
| Pages crawled | ~39 B total, ~12 B unique |
| Unique hosts | ~500 M (extrapolated) |
| Crawling effort | 1,927 machine-days |
| Avg. throughput | ~20 M URLs / day |

## Index

| | |
|---|---|
| Documents indexed | ~10.7 B |
| Unique domains | ~75 M |
| Public datasets | 511 main / 1,511 total |
| Index size | 45 TB (main) / 50 TB total |
| Embeddings | 119 M docs · 884 GB |

**Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

**Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document

**Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

# Step 1: Crawling the Web

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**OWLer — Open Web Crawler**

- Up to **105 M URLs / day** across 4 European data centers
- Central URL Frontier at **CERN** — 2 B+ URLs in crawl space
- **1.25 PiB** collected · **1,927 machine-days** · ~39.6 B fetches
- European language and domain bias
- Only HTML; No headless browsing

## Constraints

- Downstream processing caps: **1–2 TB/day per DC**
- Factor ~20× below commercial freshness
- Requires 2–4 dedicated engineers for 24/7 ops

# Step 2: Preprocessing

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### Resilipipe — WARC → Parquet Pipeline

- Two-stage pipeline: **Resilipipe (CPU)** for extraction & enrichment, **Resilipipe GPU** for embeddings & classifiers
- Modular architecture — 10+ pluggable modules (links, structured data, quality scoring, trigger warnings, …)
- Custom parallelization on HPC: SLURM + Apache Airflow orchestration
- Output: column-oriented **Apache Parquet** with per-document metadata, embeddings, and prediction scores

### Scale & Constraints

- **108 M documents** embedded with jina-embeddings-v3 (785 GB across 55 published datasets)
- GPU stage bottleneck: embedding throughput limits end-to-end pipeline speed
- Downstream cap of **1–2 TB/day per data center** inherited from crawling

### Key Enrichments

- Content extraction & language detection
- Link graph & structured data (FAQs, geo, licenses)
- Curlie taxonomy labels
- Quality scoring (ONNX)
- Trigger warnings (14 languages, 19 categories)
- Dense embeddings (jina-embeddings-v3)
- Phishing classifier & LLM detector

# Step 3: Building the Index

# Index Format: CIFF + Parquet

**CIFF (Common Index File Format)**

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately.

**Parquet Metadata Files**

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality / importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

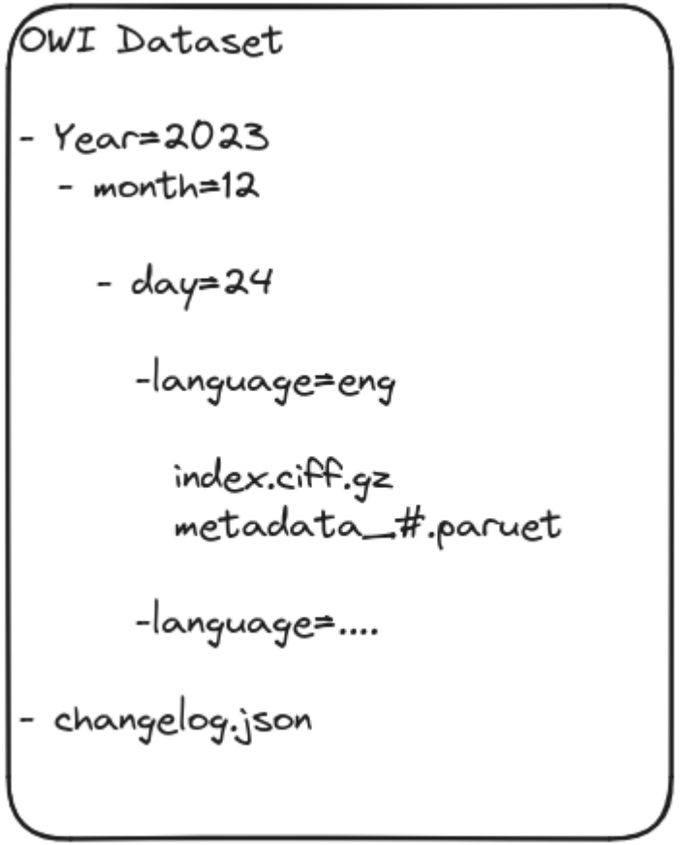
**Dense Embeddings (GPU slice)**

- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

# Dataset Structure & Versioning

Each **daily shard** follows a consistent folder hierarchy:

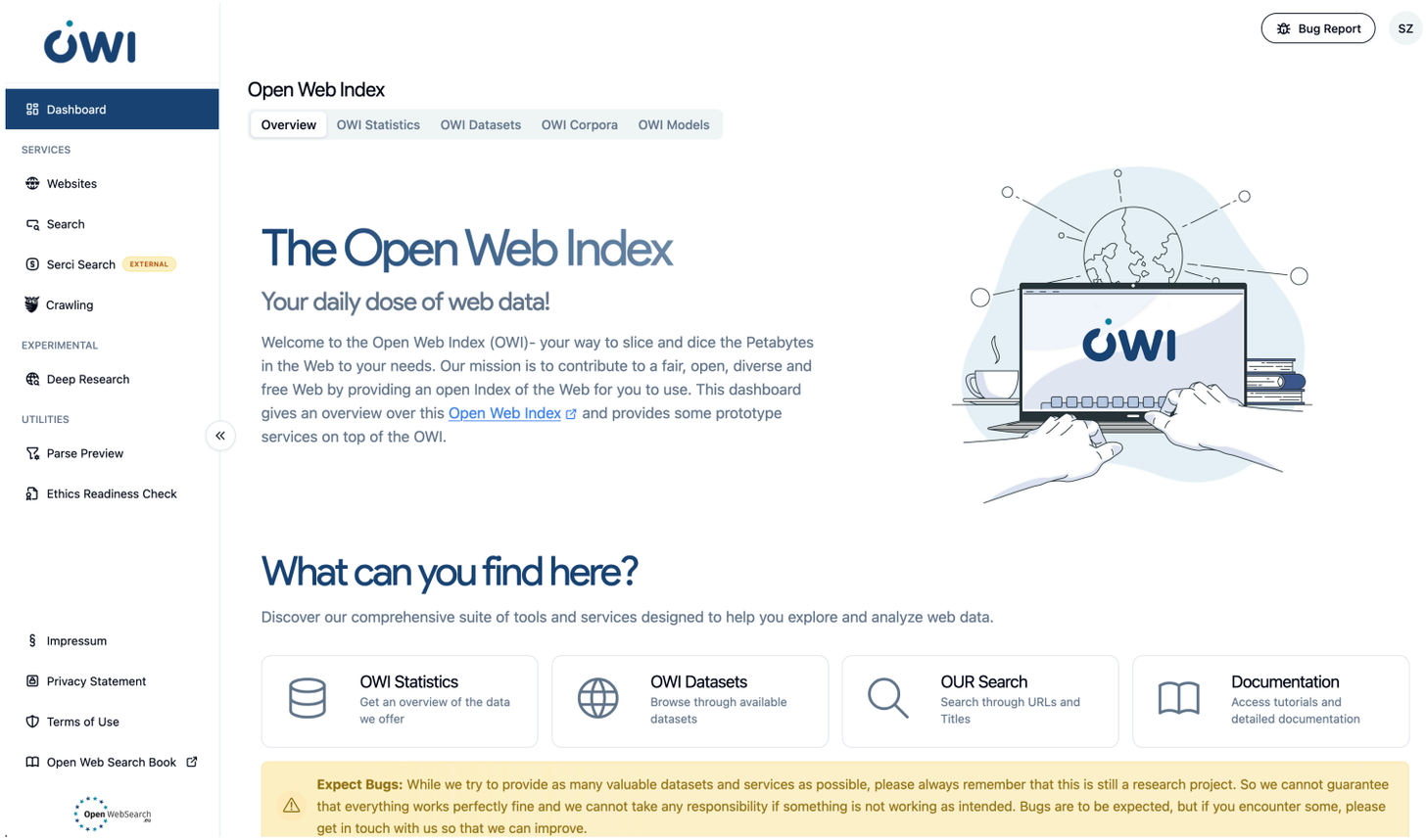
```
OWI Dataset
  └── year=YYYY
      └── month=MM
          └── day=DD
              ├── language=eng
              │   ├── index.ciff.gz
              │   └── metadata_*.parquet
              ├── language=deu
              │   └── ...
              └── changelog.json
```




- One dataset per data center per day
- Collections: `main`, `curlie_full`, `gpu` (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: `pull`, `push`, `build`

# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

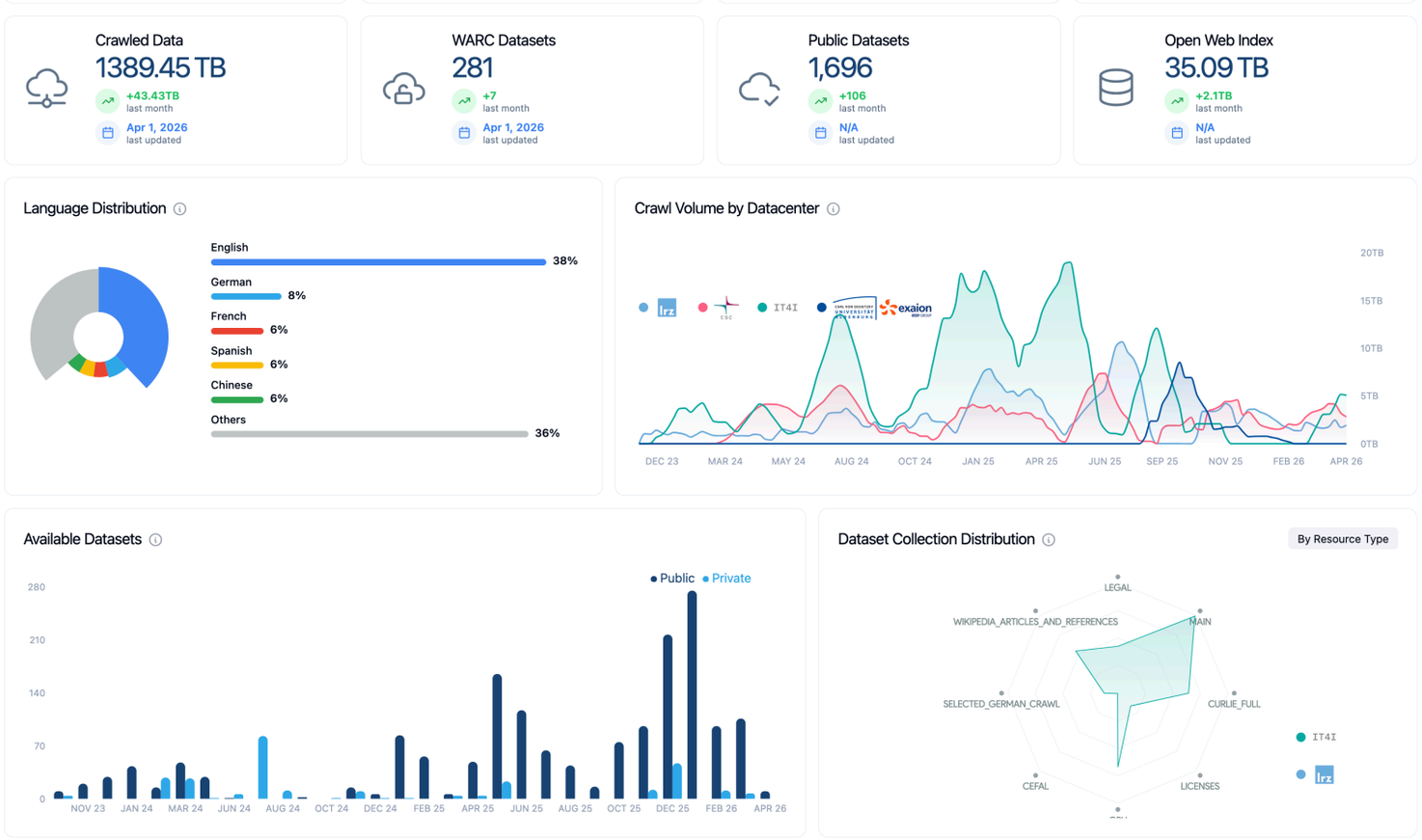



424 registered users · 17,000+ unique visitors · 70,000+ page views

**Central entry point for the Open Web Index [openwebindex.eu](openwebindex.eu)**

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

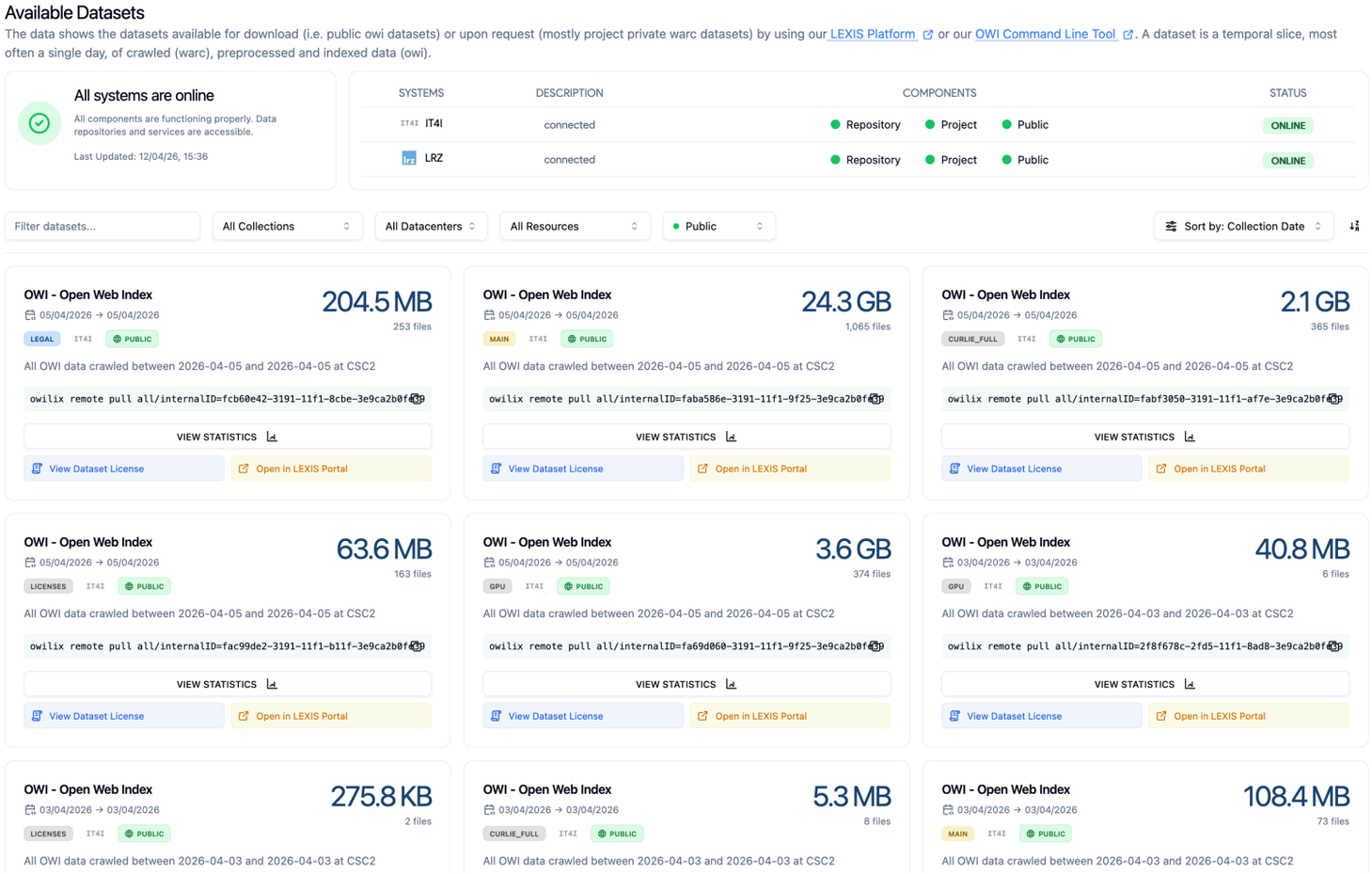
# Dashboard — Statistics & Datasets




## OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

# Dashboard — Statistics & Datasets




## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**, **structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

### OWILIX - Open Web Index CLI

version 5.3.1 | python 3.11+ | license Apache 2.0

A command-line tool for managing OpenWebIndex datasets across local and remote repositories.

#### Quick Start

##### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/

### Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

```
[→  ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
                    Search results (10 results in 113.4s)
```

| score | date | url |
|---|---|---|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/ |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/ |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/ |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/ |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/ |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/ |

## The OpenWebSearch Book

**Comprehensive Documentation at [openwebindex.eu/book](openwebindex.eu/book)**

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

## Access Modes at a Glance

| Tool | Mode | Best for |
|---|---|---|
| Dashboard | Browser | Explore, monitor |
| owilix | CLI | Download, slice, build |
| Remote index | DuckDB/SQL | Query + download |
| MosaicRAG | Conversational | RAG & chat over OWI |

> All software, data formats and access tools are **open source**

# Using the Open Web Index — Applications

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → mosaic.ows.eu
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index →

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems,

args.me/owi

- **CERN Search** — institutional search + disability knowledge search (Nooon)

and domain-specific applications

# Using the Open Web Index — Data & Analytics

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## Published Datasets

| Dataset | Scale | Use case |
|---|---|---|
| **WebFAQ** | 96M Q&A, 75 langs | QA training, cross-lingual IR |
| **WebFAQ 2.0** | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| **Imprints** | 5.25M hosts, geocoded | Enterprise analytics, Eurostat |
| **German Commons** | 154B tokens | German LLM pre-training |
| **OWS-Curlie-2025** | 20.7M docs + qrels | IR evaluation benchmark |

All datasets available at openwebindex.eu/corpora

## Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

> The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# OUR²S — Operating Open Retrieval at European Scale

# OUR²S - OWI Universal Retrieval Research System at a Glance

**Goal:** Serve the Open Web Index at Scale for Retrieval Research and Beyond.

## Stack

- **Vespa** — search, ranking, corpus-native storage
- **CockroachDB** — transactional control plane
- **Redis Streams** — event bus and cache
- **FastAPI** — REST API, admin UI, agent surface

## Design principles

1. one Vespa application, multiple corpora
2. Additional preprocessing for ranking
3. corpus isolation for ranking statistics
4. typed, model-driven
5. search-plane and control-plane separation
6. a single REST surface for admin, search, agents, and import

# Operating an OUR²S Cluster — Real Numbers

## Current Setup

- **10 bulk content nodes** — HDD-backed, document text + summaries + assets
- **8 fast content nodes** — NVMe-backed, entities + latency-critical indexes
- redundancy 1 at the Vespa layer; **Ceph** provides storage-level durability
- Vespa, CockroachDB, Redis, FastAPI, all on Kubernetes
- **Setup** centered around k8s deployemnt.

## Measured scaling laws

| Scale | Attr-RAM floor | Content storage |
|---|---|---|
| 100M pages | 55–80 GB | ~3.2 TB |
| 1B pages | 0.55–0.80 TB | ~32 TB |
| 5B pages | 2.75–4.0 TB | ~160 TB |

Planning envelope: 2x–3x measured attribute RAM for query load, feed pressure, compaction, restart margin.

> Scaling-up with uage not tested yet; Ideally, the OUR²S instances will be distributed across the federated partners.

# Agentic Search with YOARS (in development)

**Y**our **O**pen-Web Index **A**gent **R**etrieval **S**ystem — the agent-facing collaboration layer over OUR²S.

### Core idea

- Future search must be tailored for agents
- Agents need to maintain their own knowledge base of knowledge skills, sources, workflows, and tools

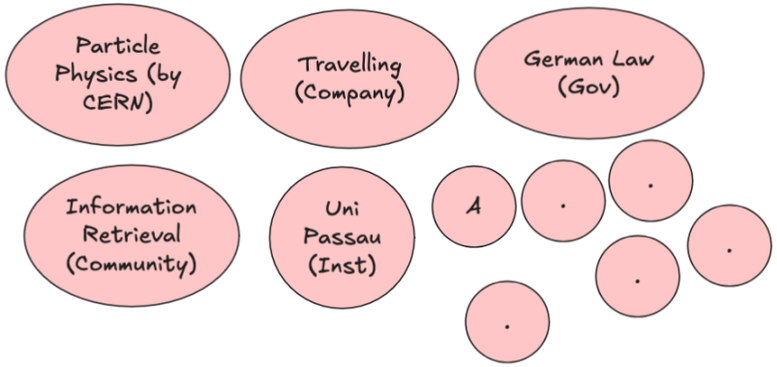
### Our Approach

- Hypermedia onboarding and API Access for Agents
- Agents create, query, and curate nuggets (atomic, retrievable, agent-maintained knowledge)
- a purposeful social network: discovery of skills, sources, workflows, tools

Open agent infrastructure on top of an open web index as possible open European AI Agent collaboration plattform.

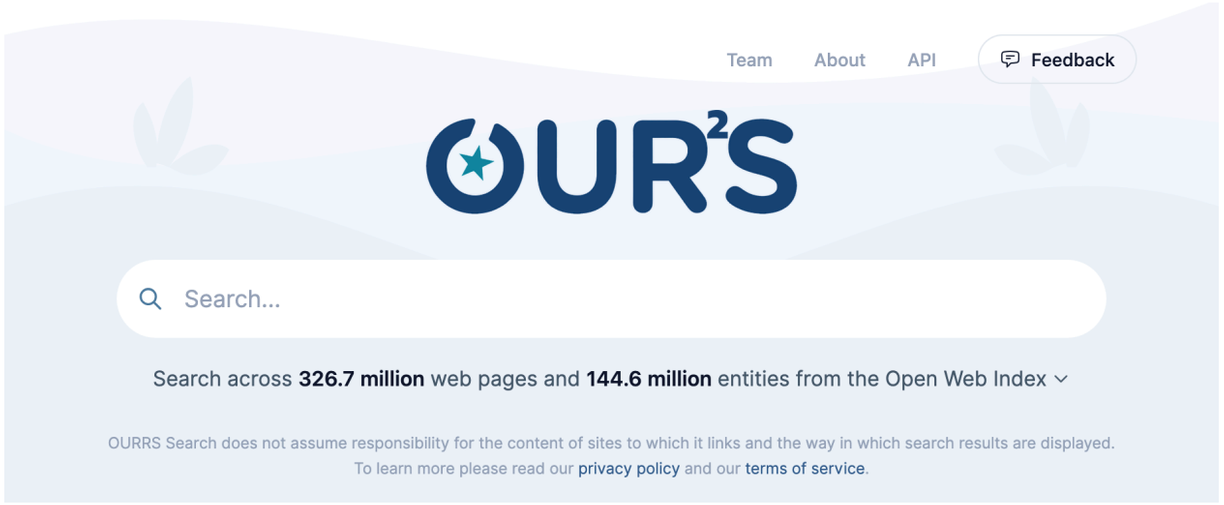
# Our Vision: Federated Web Search across Legal Entities

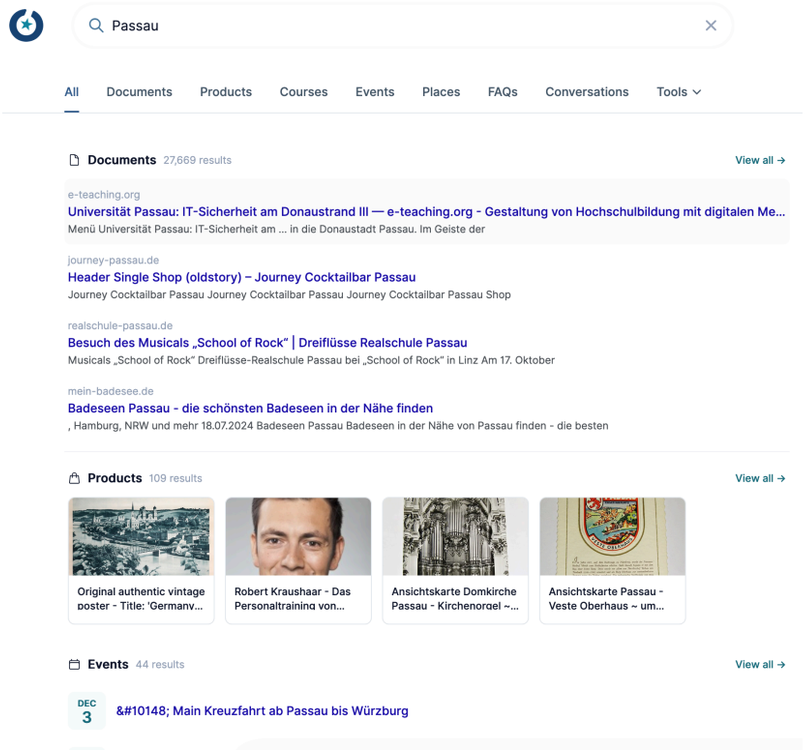
**Not another Hyperscaler:** The Open Web Index aims to reduce upfront costs for tapping the web as a resource. With OUR²S we provide a low-cost retrieval system to run in your own datacenter, which can be federated across legal entities and partitioned across topics, languages, domains, etc.
**Agentic Search** can utilise federated search systems easily, with proper guidenace and skill management which can be coordinated by YOARS!




> Can we join forces to build a European sovereign agent-based search stack that can truly compete with Hyperscalers?

# OUR²S in Action - ourrs.eu




Live, hosted, federated retrieval over the Open Web Index — open source, self-hostable, and ready for partner-operated nodes.

# Conclusion

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

| | |
|---|---|
| **Website** | openwebsearch.eu |
| **Dashboard** | openwebindex.eu |
| **Code Repository** | https://opencode.it4i.eu/openwebsearcheu-public/ |
| **Community** | openwebsearch.eu/community |
| **Join** | join@openwebsearch.org |
| **General** | community@openwebsearch.eu |

**Thank you — Questions, Comments, Feedback very welcome!**